

HW/Lab 5: MIMO Theory

ECSE 4560/6560: Modern Communication Systems

RPI Fall 2025

Goals and Instructions

The goals for this assignment are:

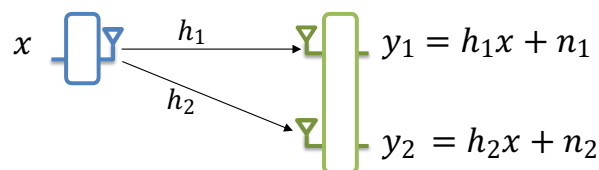
- Model single-stream MIMO/SIMO/MISO links and compute instantaneous SNRs and throughput.
- Reason about diversity gain, array gain, and their effect on reliability and throughput based on a practical IEEE 802.11 WiFi example.
- Understand channel conditions favorable to support MIMO multiplexing.

Follow these instructions:

- **Collaboration:** You can discuss concepts, but write your own solutions and code.
- The exam has 4 theoretical questions and one programming assignment.
- **Tools:** We will primarily use MATLAB (and/or Python). Directions use standard MATLAB functions. *Strongly recommended* toolboxes: Signal Processing, Communications. (Base-MATLAB fallbacks are acceptable.)
- **Submission:** Upload a single PDF with your answers and plots to Gradescope before the deadline.
- Total 20 points for ECSE 4560 and 25 points for ECSE 6560 students.

Question 1: Receive Diversity with MRC

Setup. A single-antenna transmitter sends complex baseband symbols $x[k]$ over a narrowband flat-fading channel to a two-antenna receiver.



The received samples are

$$y_1[k] = h_1 x[k] + n_1[k], \quad y_2[k] = h_2 x[k] + n_2[k],$$

where $h_1, h_2 \in \mathbb{C}$ are constant over a block and known at the receiver. Noises are independent $n_1, n_2 \sim \mathcal{CN}(0, N_0)$. The symbol power is $\mathbb{E}[|x[k]|^2] = P$.

- (a) **Vector model (1 line).** Write $\mathbf{y}[k] = \mathbf{h}x[k] + \mathbf{n}[k]$ by defining $\mathbf{y}[k]$, $\mathbf{h}$, and $\mathbf{n}[k]$.
- (b) **MRC decision statistic.** We use a linear combiner $\hat{x}[k] = \mathbf{w}^H \mathbf{y}[k]$. State the maximum-ratio combining (MRC) weights $\mathbf{w}$ (up to a nonzero scalar) and write the decision statistic $\hat{x}$ explicitly in terms of h_1, h_2, y_1, y_2 .
- (c) **Post-combiner SNR.** Show that the instantaneous output SNR with MRC is

$$\gamma_{\text{MRC}} = \frac{(|h_1|^2 + |h_2|^2)P}{N_0}.$$

(You may assume the standard AWGN definition $\gamma = \frac{\text{signal power}}{\text{noise power}}$ at the combiner output.)

- (d) **Numerical example.** Let

$$h_1 = 1 + j, \quad h_2 = 0.5 - j, \quad P = 1, \quad N_0 = 0.5.$$

Compute:

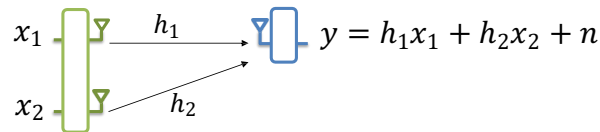
- (i) $|h_1|^2$, $|h_2|^2$, and γ_{MRC} .
- (ii) The single-branch SNRs $\gamma_1 = \frac{|h_1|^2 P}{N_0}$ and $\gamma_2 = \frac{|h_2|^2 P}{N_0}$.
- (e) **Selection combining (compare).** Selection combining chooses the receive branch with the highest instantaneous SNR and ignores the others, and detects using that single branch. Provide an expression for selection combining SNR γ_{SC} and compute the SC gain using the same numbers above. Finally, compare the gain of MRC over SC in dB: $10 \log_{10}(\gamma_{\text{MRC}}/\gamma_{\text{SC}})$
- (f) **Quick BER check (BPSK).** For coherent BPSK detection on the combiner output, the instantaneous bit error probability is

$$P_b = Q(\sqrt{2\gamma}).$$

Compute P_b using (i) $\gamma = \gamma_{\text{MRC}}$ and (ii) $\gamma = \gamma_{\text{SC}}$ from parts (d)–(e). (No need to evaluate $Q(\cdot)$ numerically; leave answers in $Q(\cdot)$ form.)

Question 2: Transmit Diversity MRC Beamforming vs. Alamouti

Setup. Two transmit antennas, one receive antenna.



Consider narrowband flat fading with channel vector

$$\mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} \in \mathbb{C}^2, \quad n \sim \mathcal{CN}(0, N_0), \quad \mathbb{E}[|x|^2] = P.$$

Unless stated otherwise, total TX power per channel use is P (shared across the two antennas).

- (a) **MRC beamforming with CSIT (derivation).** With perfect Channel state information at the transmitter (CSIT), the transmitter applies a unit-norm precoder $\mathbf{w} \in \mathbb{C}^2$ and sends $\mathbf{x} = \mathbf{w}x$.

(i) Under $\|\mathbf{w}\|_2 = 1$, derive $\mathbf{w}$ that maximizes instantaneous receive SNR.

(ii) Show that the resulting SNR equals $\gamma_{\text{BF}} = \frac{\|\mathbf{h}\|_2^2 P}{N_0} = \frac{(|h_1|^2 + |h_2|^2)P}{N_0}$.

- (b) **naïve repeat without CSIT.** Suppose we *ignore* CSIT and simply send the same symbol from both antennas with equal power: $\mathbf{x} = \frac{1}{\sqrt{2}}[1 \ 1]^\top x$.

(i) Write the received signal $y = (h_1 + h_2)\frac{1}{\sqrt{2}}x + n$ and derive the corresponding SNR γ_{rep}

(ii) State one condition on (h_1, h_2) under which γ_{rep} becomes *much smaller* than γ_{BF} (brief, one line).

- (c) **Alamouti STBC (no CSIT).** Using the 2×1 Alamouti code over two time slots:

$$\text{slot } t: [x_1 \ x_2], \quad \text{slot } t+1: [-x_2^* \ x_1^*].$$

(i) Write the two received samples y_t and y_{t+1} in terms of $h_1, h_2, x_1, x_2, n_t, n_{t+1}$.

(ii) Form the standard two-equation linear combiner that decouples (x_1, x_2) and show that each symbol experiences an effective gain $|h_1|^2 + |h_2|^2$ (assume total TX power per *time slot* equals P and the usual Alamouti normalization).

(iii) State the resulting per-symbol SNR γ_{Alam} in terms of $P, N_0, |h_1|^2, |h_2|^2$ and the code rate.

- (d) **Numerical comparison (use the same channel for all parts).** Let

$$h_1 = 0.6 + 0.8j, \quad h_2 = -0.3 + 0.4j, \quad P = 1, \quad N_0 = 0.2.$$

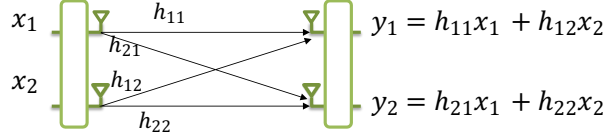
(i) Compute $|h_1|^2$, $|h_2|^2$, and $\|\mathbf{h}\|_2^2$.

(ii) **CSIT beamforming:** compute γ_{BF} and express it also in dB.

(iii) **naïve repeat:** compute γ_{rep} and the loss (dB) relative to γ_{BF} .

(iv) **Alamouti:** compute γ_{Alam} and compare (dB) to γ_{BF} .

- (e) **Explanation.** When CSIT is unavailable or unreliable, explain in one or two sentences why Alamouti is preferred over naïve repetition for reliability at the same total transmit power.



Question 3: Simple 2×2 MIMO with Zero-Forcing

Setup. Consider a narrowband flat-fading 2×2 MIMO link for a single channel use:

$$\mathbf{y} = \mathbf{H} \mathbf{x} + \mathbf{n}, \quad \mathbf{H} \in \mathbb{C}^{2 \times 2} \text{ invertible},$$

with $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $\mathbb{E}[|x_1|^2] = \mathbb{E}[|x_2|^2] = P$, and $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I}_2)$. The receiver knows $\mathbf{H}$ perfectly.

- (a) **ZF equalizer and effective noise (one-step derivation).** Use the zero-forcing equalizer $\mathbf{W} = \mathbf{H}^{-1}$. Show that

$$\hat{\mathbf{x}} = \mathbf{W} \mathbf{y} = \mathbf{x} + \mathbf{H}^{-1} \mathbf{n}.$$

Write the two decision statistics explicitly:

$$\hat{x}_1 = [\mathbf{H}^{-1}]_{1,:} \mathbf{y}, \quad \hat{x}_2 = [\mathbf{H}^{-1}]_{2,:} \mathbf{y},$$

and identify the corresponding effective noise terms on each stream.

- (b) **Numeric evaluation.** Let

$$\mathbf{H} = \begin{bmatrix} 1 & 0.5 \\ -0.3 & 1.2 \end{bmatrix}, \quad P = 1, \quad N_0 = 0.2.$$

Compute:

- (i) $\mathbf{H}^{-1}$ (show your intermediate steps briefly).
- (ii) The squared row norms of $\mathbf{H}^{-1}$: $\|\mathbf{w}_1\|^2 = \|[\mathbf{H}^{-1}]_{1,:}\|^2$, $\|\mathbf{w}_2\|^2 = \|[\mathbf{H}^{-1}]_{2,:}\|^2$.
- (iii) The post-ZF SNRs using these norms (no extra matrices):

$$\gamma_1 = \frac{P}{N_0 \|\mathbf{w}_1\|^2}, \quad \gamma_2 = \frac{P}{N_0 \|\mathbf{w}_2\|^2}.$$

- (iv) Which stream is more noise-enhanced? State your answer and the SNR ratio γ_1/γ_2 in dB.

- (c) **Explanation.** When does ZF incur severe performance degradation through noise enhancement in 2×2 MIMO? State a condition on the channel matrix $\mathbf{H}$.

Question 4: (ECSE 6560 only) MIMO Throughput for an 802.11n-like Link

Setup. One OFDM *stream* over a 20 MHz channel with 64-point FFT, $N_{\text{data}} = 52$ payload subcarriers, OFDM symbol duration $T_s = 4 \mu\text{s}$ (OFDM symbol rate = 250,000 symbols/s). The receiver has perfect CSI. Compare two *single-stream reliability* modes:

$$\text{SISO } (1 \times 1), \quad \text{SIMO receive diversity } (1 \times 2 \text{ MRC}).$$

Assume total transmit power $P = 10$ and noise $N_0 = 0.1$ (per complex sample). Use SNR in dB throughout.

SNR (dB) is given by:

$$\gamma_{\text{SISO,dB}} = 10 \log_{10} \left(\frac{|h|^2 P}{N_0} \right), \quad \gamma_{\text{MRC,dB}} = 10 \log_{10} \left(\frac{(|h_1|^2 + |h_2|^2) P}{N_0} \right).$$

MCS table: Practical links pick the highest modulation/coding scheme (MCS) whose *threshold in dB* is not above the instantaneous SNR in dB, ensuring a target reliability.

How to use it (simple rule in dB):

- Compute $\gamma_{\text{SISO,dB}}$ or $\gamma_{\text{MRC,dB}}$.
- Pick the *highest* MCS whose threshold (dB) satisfies $\gamma_{\text{dB}} \geq \gamma_{\text{th,dB}}$.

MCS thresholds (instantaneous SNR, dB):

MCS	$\gamma_{\text{th,dB}}$
BPSK, $R = \frac{1}{2}$	3
QPSK, $R = \frac{1}{2}$	6
QPSK, $R = \frac{3}{4}$	10
16-QAM, $R = \frac{1}{2}$	14
16-QAM, $R = \frac{3}{4}$	18
64-QAM, $R = \frac{1}{2}$	22
64-QAM, $R = \frac{3}{4}$	26
256-QAM, $R = \frac{1}{2}$	30
256-QAM, $R = \frac{3}{4}$	34

Bits per OFDM symbol (single stream):

$$\text{bits/symbol} = 52 \times b_{\text{mod}} \times R, \quad b_{\text{mod}} \in \{1, 2, 4, 6, 8\} \text{ for BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM.}$$

Counts data subcarriers and error correcting codes via R . Pilots/MAC overhead ignored.

Throughput: Throughput = (bits/symbol) × (symbols/s). With $1/T_s = 250,000$:

$$\text{throughput (Mbps)} = \frac{\text{bits/symbol} \times 250,000}{10^6}.$$

Single channel scenario (use exactly these values, with $h := h_1$):

$$h = h_1 = 0.6 + 0.2j, \quad h_2 = 0.4 - 0.8j.$$

- Compute SNRs (dB) and choose MCS.** For each mode (SISO, MRC), compute γ_{dB} using the formulas above and select the highest feasible MCS from the dB table.
- Throughput.** Compute throughput for both SISO and MRC. Also report the improvement ratio = (MRC throughput) / (SISO throughput).
- Explanation.** Explain the throughput improvement of SIMO MRC over SISO.
- Explanation.** What would happen in terms of effective throughput and bit error rate if the MCS is chosen wrongly, discuss both cases when MCS is chosen a smaller value and a larger value in the table for a given SNR. Does the throughput increase or decrease? Does the bit error rate increase or decrease? Explain your answer intuitively (no calculation needed).

Question 5: Programming: Rank, Invertibility, and SVD (2×2)

In Zero-forcing detection, noise amplification is governed by the *singular values* of $\mathbf{H}$ (from $\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$). Singular values are real and non-negative: $\mathbf{\Sigma} = \text{diag}(\sigma_1, \sigma_2)$ with $\sigma_1 \geq \sigma_2 \geq 0$.

You are given the following channel matrices

$$\mathbf{H}_1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{H}_2 = \begin{bmatrix} 1 & 0.6 \\ -0.2 & 1.1 \end{bmatrix}, \quad \mathbf{H}_3 = \begin{bmatrix} 0 & 1 \\ 1 & 0.1 \end{bmatrix}.$$

- (a) **Singular values, rank, invertibility.** For each $\mathbf{H}_k$, report (σ_1, σ_2) with $\sigma_1 \geq \sigma_2$, the matrix rank, the determinant, and state if the channel matrix is *Invertible?* (Yes/No).
Hint: invertible $\Leftrightarrow \sigma_2 > 0 \Leftrightarrow \det(\mathbf{H}_k) \neq 0$.
- (b) **Condition number and ZF sensitivity.** For each *invertible* $\mathbf{H}_k$, report $\kappa_2 = \sigma_1/\sigma_2$ and classify: *well-conditioned* (close to 1) vs *poorly conditioned* (large). One sentence explaining how large κ_2 impacts ZF noise.
- (c) **Explanation.** When $\sigma_{\min} \rightarrow 0$, which is safer to use at the same total power (i) ZF spatial multiplexing or (ii) single-stream diversity (TX diversity/RX MRC)? Give one reason tied to $\sigma_{\min}$ and ZF noise amplification.

Report format (single PDF). Submit **one PDF** exported from a MATLAB or Python that shows your code *and* outputs *and* brief answers.

Allowed tools (use any of these). *MATLAB:* `svd`, `rank`, `det`, `cond` (2-norm by default). *Python (NumPy):* `numpy.linalg.svd`, `matrix_rank`, `det`, `cond`. (You may also use `scipy.linalg.svd` etc. Results should be rounded to two decimals.). This is a simple code to write, so there is no helper code provided.